

Reviewer Pack — Minimal Rank of Real Algebraic Geometry Invariants vs Sum-of...

Entry 4e9e9d0e9688 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Real Algebraic Geometry Invariants vs Sum-of-Squares Degree for Max-CUT

Entry ID: 4e9e9d0e9688 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-25 19:46:01 UTC

1. Conjecture

Field A (mathematical branch): Real algebraic geometry **Field B** (complexity object): Complexity Theory: Sum-of-squares degree for max-CUT

Statement:

[‘For any SOS degree-d polynomial that approximates max-CUT with a ratio better than 0.878, the real algebraic geometry invariant of its moment matrix must have a rank at least d.’, ‘Any max-CUT approximation polynomial with an SOS degree-d less than or equal to d has a moment matrix whose real algebraic geometry invariant’s rank is exactly d.’]

Rationale (proposer’s reasoning):

[‘Real algebraic geometry invariants provide a rich structure for analyzing the complexity of computational problems. If such invariants can be linked to the SOS degree for max-CUT, it would offer new insights into understanding the hardness of this problem.’, ‘The connection between real algebraic geometry and complexity theory could potentially lead to novel algorithms or prove lower bounds for problems like max-CUT.’]

Taxonomy category: SOS_DEGREE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 9af04adb7d162cc5

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if for all max-CUT instances with $n \leq 40$ variables, the SOS degree-d polynomial approximation has a ratio better than 0.878 and its moment matrix’s real algebraic geometry invariant rank is at least d. The conjecture is falsified if there exists any instance where the approximation fails to meet these conditions.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	UNCERTAIN	0.90	UNCERTAIN	HITS
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = 40
    instances_tested = 30

    def generate_max_cut_instance(n):
        edges = []
        for i in range(n):
            for j in range(i + 1, n):
                if random.random() < 0.5:
                    edges.append((i, j, random.random()))
        return edges

    def construct_moment_matrix(edges):
        M = [[0] * (n + 1) for _ in range(n + 1)]
        for i in range(n):
            M[i][i] = 1
        for u, v, w in edges:
            M[u][v] += w
            M[v][u] += w
        return M

    def gaussian_elimination(M):
        n = len(M)
        rank = 0
        for i in range(n):
            if M[i][i] == 0:
                swapped = False
                for j in range(i + 1, n):
```

```

        if M[j][i] != 0:
            M[i], M[j] = M[j], M[i]
            swapped = True
            break
    if not swapped:
        continue
    pivot = M[i][i]
    for j in range(n + 1):
        M[i][j] /= pivot
    for j in range(n):
        if j != i and M[j][i] != 0:
            factor = M[j][i]
            for k in range(n + 1):
                M[j][k] -= factor * M[i][k]
    rank += 1
return rank

def sos_degree_polynomial(edges, d):
    # Placeholder function to simulate SOS degree calculation
    return random.randint(0, d)

total_ratio = 0
for _ in range.instances_tested):
    edges = generate_max_cut_instance(n)
    M = construct_moment_matrix(edges)
    rank = gaussian_elimination(M)
    d = sos_degree_polynomial(edges, n)

    if rank < d:
        return {
            "metric_name": "Rank of Moment Matrix",
            "metric_value": rank,
            "instances_tested": instances_tested,
            "conjecture_holds": False,
            "counterexample": f"Ratio: {total_ratio / instances_tested}, Rank: {rank}"
        }

    total_ratio += max_cut_approximation(edges)

return {
    "metric_name": "Rank of Moment Matrix",
    "metric_value": rank,
    "instances_tested": instances_tested,
    "conjecture_holds": True,
    "counterexample": ""
}

def max_cut_approximation(edges):
    # Placeholder function to simulate max-cut approximation
    return random.random() * 0.878

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else [2, 3, 5, 7, 11, 13, 17, 19, 23, 29]

    results = []

```

```

for seed in seeds:
    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

mean_value = sum(r["metric_value"] for r in results) / len(results)
std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
elif any(not r["conjecture_holds"] for r in results):
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"Ratio too low\" first_failing_seed={first_failing_seed}")
else:
    print("RESULT: INCONCLUSIVE reason=unknown")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_57ffdc6e.py", line 105, in <module>
    result = run_trial(seed)
              ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_57ffdc6e.py", line 73, in run_trial
    rank = gaussian_elimination(M)
           ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_57ffdc6e.py", line 56, in gaussian_elimination
    M[i][j] /= pivot
    ~~~~^
IndexError: list index out of range

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing any data, which means we cannot verify the conjecture's conditions. | next: Re-run the test with appropriate error handling to ensure it completes without crashing.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11224
2	preregistration	ollama_remote	glm4:latest	0	0	5773
3	novelty	ollama_remote	glm4:latest	0	0	5688
4	novelty	ollama_remote	glm4:latest	0	0	5219
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14758
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12526
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11201
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11468
9	judge	ollama_remote	glm4:latest	0	0	11802

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 89659 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/4e9e9d0e9688.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/4e9e9d0e9688.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/4e9e9d0e9688.tar.gz` (if generated)